



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO THANGGAMANI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

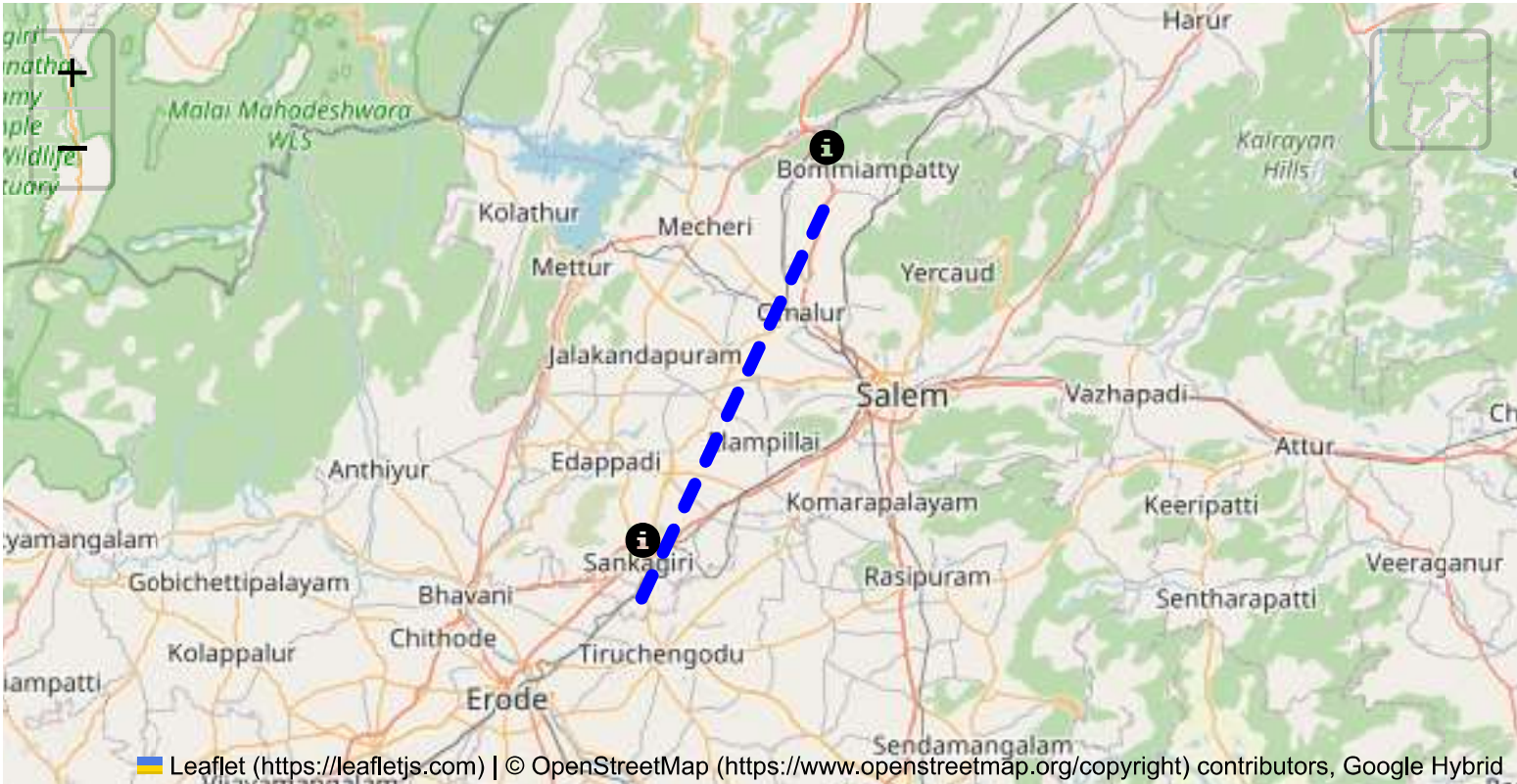
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

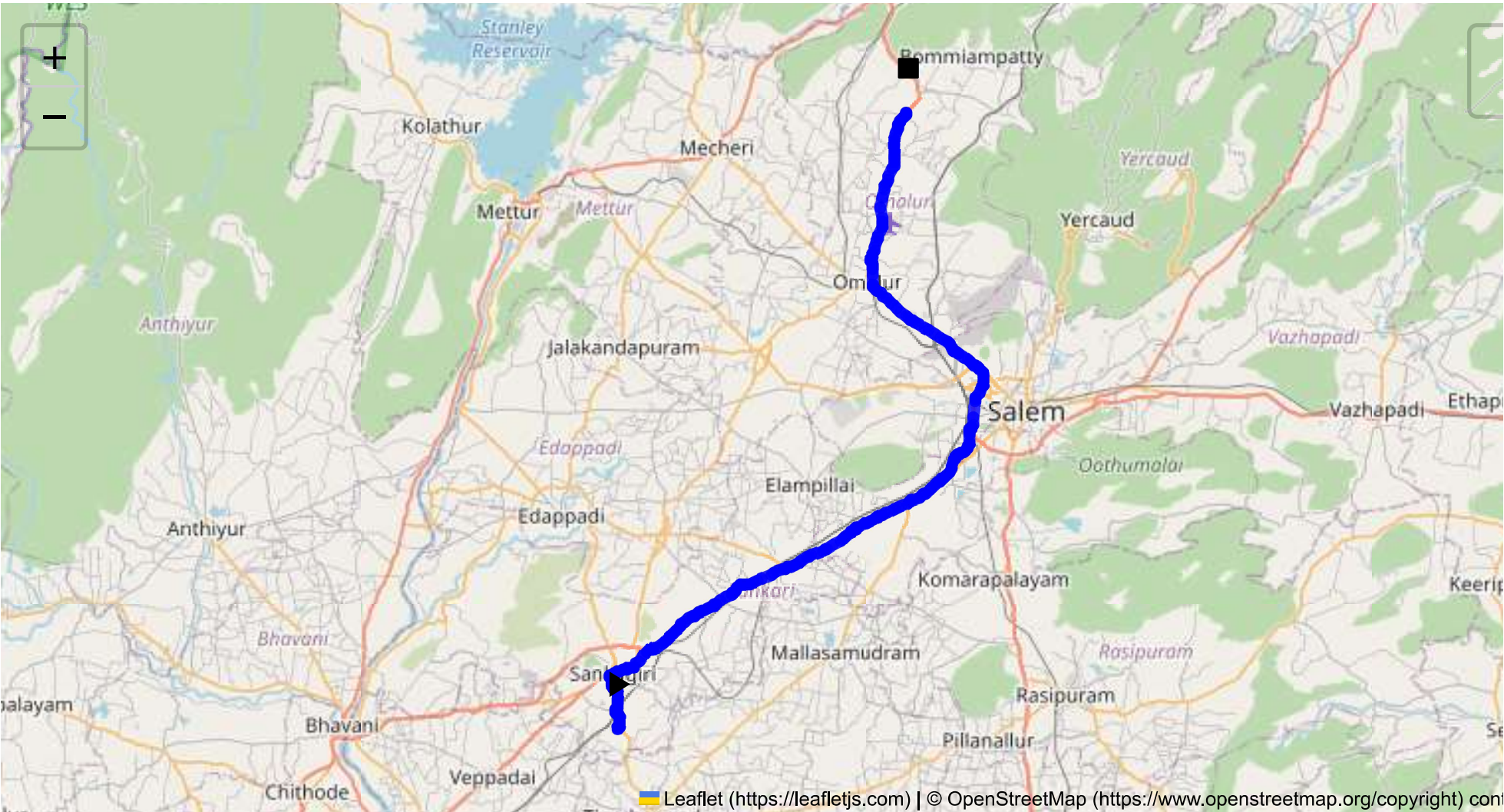
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 68.03 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.857676, 78.074142)



Welcome to the Journey Risk Management Study

Route Overview:

The route from Sangagiri to Salem in Tamil Nadu primarily navigates through a series of highways and regional roads. This transit extends approximately 68.03 kilometers and typically requires about 1.39 hours for heavy vehicles carrying hazardous materials. The journey includes passage through rural zones with certain urban centers, increasing vehicle interactions and infrastructure variations.

Weather Conditions:

- **Typical Conditions:** The region generally experiences a tropical climate with a dry season from January to March, a hot summer from April to June, and a rainy monsoon from July to October.
- **Potential Hazards:** Heavy rainfall during the monsoon could lead to reduced visibility, water-logging, and slippery road conditions. Dust during the dry season might impact visibility and the truck's air filtration systems.

Traffic Patterns:

- **Peak Hours:** Morning (7 AM to 9 AM) and evening (5 PM to 8 PM) records the highest traffic due to commuter transitions and school transport activities.
- **Congestion-prone Areas:** Slight congestion near urban outskirts, particularly as the route approaches Salem city.

Road Quality and Infrastructure:

- **Quality:** Mixed, with national highways presenting high-quality, well-maintained conditions, while some regional roads exhibit wear and may have potholes and uneven surfaces, particularly after the monsoon.
- **Infrastructure:** Adequate signaling, but occasional narrow passages in less urbanized areas could create bottlenecks or complicate passing maneuvers.

Alternative Routes:

In emergencies, consider using liaison roads between major highways, such as the NH44, which runs parallel and serves the region extensively. This alternative offers more consistent infrastructure and clearer emergency facilities.

Local Regulations on Hazardous Material Transport:

- Strict adherence to weight limits and transport hours for hazardous materials is mandatory.
- Required documentation and advance notice for transporting dangerous goods are essential, especially through urban zones.

Historical Incidents:

- **Heavy Vehicles:** Few minor incidents often attributed to wet conditions and human error.
- **Hazardous Materials:** No significant incidents reported recently. Nevertheless, adherence to safety protocols is critical.

Environmental Considerations:

- **Sensitive Areas:** Proximity to agricultural zones requires careful emission management and spillage prevention.

- **Natural Habitats:** Preservation efforts nearby necessitate cautious driving to avoid disturbances, especially in rural ecological niches.

Communication Coverage:

- **Dead Zones:** Sporadic coverage gaps in rural stretches. Networks tend to improve near urban centers like Salem.
- **Recommendations:** Equip vehicles with satellite communication devices for emergencies.

Emergency Response Times:

Response times can vary but expect approximately 20-30 minutes nearer to Salem due to urban response units. Rural sections might see times extend to 45 minutes due to limited access and personnel availability.

Overall Risk Assessment:

This route presents a moderate-risk level under normal conditions. Concerns primarily revolve around weather impacts, road quality, and peak-hour traffic congestions. Driver awareness and compliance with safety regulations are crucial to mitigating risks effectively. Emergency response preparedness and communication equipment will bolster safety measures.

Overall, thorough route planning with contingencies considered for adverse conditions and community regulations will ensure efficient and safe journey completion for hazardous materials transport along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.45004, 77.87206	15 KM/Hr
4	Turn	Medium	11.45028, 77.87202	30 KM/Hr
5	Turn	Medium	11.45309, 77.87325	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.61736, 78.10588	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.61753, 78.10608	30 KM/Hr
11	Turn	Medium	11.64007, 78.11794	30 KM/Hr
12	Turn	Medium	11.64020, 78.11786	30 KM/Hr
13	Turn	High	11.69390, 78.10964	15 KM/Hr
14	Turn	High	11.69394, 78.10964	15 KM/Hr
15	Turn	Medium	11.85709, 78.07424	30 KM/Hr
16	Turn	High	11.85747, 78.07423	15 KM/Hr
17	Turn	High	11.85766, 78.07454	15 KM/Hr
18	Turn	High	11.85774, 78.07450	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
24	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
25	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
26	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
27	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
29	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
30	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
31	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
34	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
35	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
37	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
38	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
39	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
40	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
42	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
28	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
32	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
33	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
36	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
41	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45004, 77.87206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64007, 78.11794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64020, 78.11786



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69390, 78.10964



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69394, 78.10964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.85709, 78.07424



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.85747, 78.07423



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.85766, 78.07454



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.85774, 78.07450

